

# Online Learning-Induced Port Selection for Fluid Antenna in Dynamic Channel Environment

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**Abstract**—To expand the degrees of freedom and achieve additional gains beyond multiple-input multiple-output (MIMO), the fluid antenna system (FAS) has been regarded as a potential approach for exploiting spatial diversity by reconfigurable position-switchable antennas. In the operation of FAS, port selection is one of the key challenges, demanding substantial channel training overhead. In this letter, we propose a novel online learning framework to address the port selection issue without the instantaneous full channel state information (CSI) in time-varying channel conditions. Based on bandit learning, the proposed approach learns the optimal port selection by dynamically interacting with the varying channels. The proposed algorithm is proved to achieve guaranteed performance with a bounded deviation and simulation results confirm comparable performance by comparing with the ideal optimal selections.

**Index Terms**—Fluid antenna systems, port selection, bandit learning.

## I. INTRODUCTION

MULTIPLE-INPUT multiple-output (MIMO) technology has been regarded as a crucial part of modern wireless networks due to its multiplexing and diversity gain. It has been widely studied during the last two decades and applied to many wireless standards, shifting from multiuser MIMO in the fourth generation (4G) to massive MIMO in the fifth generation (5G) networks [1]. Traditional MIMO techniques not only need channel state information (CSI) but also require high-complexity signal processing operations to generate an orthogonal beamformer for realizing multiuser multiplexing. However, MIMO, especially for massive MIMO, is suitable for base station (BS), which cannot be applied to the user terminal due to the limited space. Therefore, the user terminal with poor capability becomes the performance bottleneck of wireless networks. Recently, the proposed fluid antenna offers a promising solution to the dilemma [2].

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The fluid antenna is a new reconfigurable antenna that has no pre-defined form and can be adjusted to the desired shape through changing antenna position (also referred to as ‘port’). It can be fabricated using fluidic materials or switchable pixels [3]. Despite the fluid antenna has received increasing attention in the antenna community during the past decade, it was not until the work of [4] investigated the potential of applying fluid antenna to improve wireless communication performance. Recently, the work of [5] exploited the spatial diversity provided by switchable antenna port to propose a novel fast fluid antenna multiple access (*f*-FAMA). In particular, the fluid antenna allows the user terminal to adapt the port to exploit the interference null created by randomly superimposed signals and multipath fading to achieve massive connectivity. Accordingly, *f*-FAMA simplifies multiple access into a simple task of port searching at the user terminal without sophisticated signal processing at the transmitter side. However, *f*-FAMA is performed on a symbol-by-symbol basis with high complexity. To alleviate the implementation complexity, Wong et al. further proposed slow fluid antenna multiple access (*s*-FAMA) [6], which only exploits the random fading channel to find the port where all the interfering channels are in deep fading. However, the implementation of *s*-FAMA needs the CSI of all the ports, whose system overhead is intolerable due to the large number of available ports. For handling this, the works of [7], [8] adopted deep learning approaches to develop the “predict and optimize” paradigm for implementing *s*-FAMA, which predicted the full CSI based on partial port observations. Nevertheless, the generalization of deep learning hinders the wide application of such a learning-induced *s*-FAMA.

In this letter, rather than depending on fully available and instantaneous CSI, we pose the following questions:

- Can the Fluid Antenna System learn to select an optimal port without having the instantaneous full CSI available?
- Can the port selection adapt to the time-varying channel environment without periodic channel training?

To handle the above problems, through exploiting the multi-armed bandit [9], we adopt the exponential-weight algorithm for exploration and exploitation (EXP3) approach [10] to propose an online learning framework for the port selection in fluid antenna systems. In particular, We model the user terminal and the available ports as the *player* and the *arms*, respectively. Then, by utilizing bandit learning, a sequential selection is made from a set of arms, i.e., the available ports at each iteration. After drawing the selected arm, the user receives rewards for its decisions. Note that the reward is

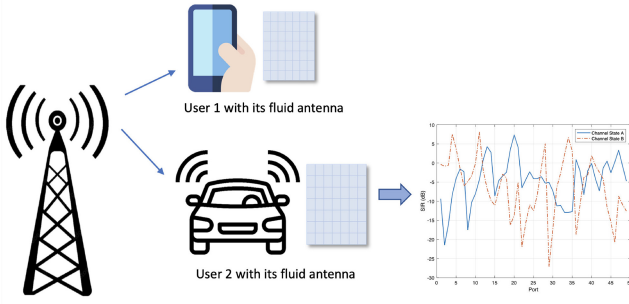


Fig. 1. System model.

dominated by unknown CSI. The contributions of the work are summarized as follows.

- We present a learning framework to learn the optimal port selection based on the theory of multi-armed bandit for the first time, in order to maximize system performance.
- We give the theoretical analysis and extensive performance evaluation of the proposed algorithm.
- We implement and evaluate the online learning algorithm to evaluate the performance over varying channels. Simulation results demonstrate that the proposed algorithm achieves comparable performance with the ideal case with perfectly known conditions.

## II. SYSTEM MODEL

We consider the downlink multiuser transmission where an BS equipped with  $N_t$  antennas broadcasts signals to  $K$  users. For simplicity, we assume that  $K = N_t$  and each communication users (CUs) is equipped with a 2D fluid antenna including  $N_r = N_{r1} \times N_{r2}$  available ports distributed in a grid structure. The area of fluid antenna  $W_r = \lambda W_{r1} \times \lambda W_{r2}$  where its horizontal and vertical space length are  $\lambda W_{r1}$  and  $\lambda W_{r2}$ , and  $\lambda$  denotes the wavelength of the carrier frequency. Besides, the BS broadcasts  $K$  data streams to users without using precoder, and each CU adopts  $s$ -FAMA where one port is activated opportunistically to maximize the instantaneous signal-to-interference ratio (SIR) for realizing multiuser multiplexing.

We adopt the MIMO fluid antenna channel model [11, eq. (3)] to build the channel from the BS to the  $k$ th CU as follows

$$\mathbf{H}_k = \mathbf{U}\sqrt{\mathbf{V}}\mathbf{G}_k, \quad (1)$$

where  $\mathbf{G}_k \in \mathbb{C}^{N_r \times N_t}$  is the small-scale fading matrix with each element as an independent zero-mean complex Gaussian random variable. Besides, the spatial correlation matrix at the receiver is  $\mathbf{J} \in \mathbb{C}^{N_r \times N_r}$ , given by [11, eq. (1)], and the eigenvalue decomposition of  $\mathbf{J}$  is  $\mathbf{J} = \mathbf{U}\mathbf{V}\mathbf{U}^T$ , where the columns in  $\mathbf{U}$  are the eigenvectors of  $\mathbf{J}$  and the diagonal entries in  $\mathbf{V}$  are the corresponding eigenvalues.

Accordingly, the received signal at the  $i$ -th port of the  $k$ -th user can be denoted as

$$r_{i,k} = s_k h_k^{(i,k)} + \sum_{j=1, j \neq k}^{N_t} s_j h_k^{(i,j)} + \sigma_{i,k}, \quad (2)$$

where  $h_k^{(i,j)}$  is the  $(i,j)$ -th element in  $\mathbf{H}_k$ , denoting the channel from the  $j$ -th antenna of the BS to the  $i$ -th port of the  $k$ -th CU.  $\sigma_k$  represents the additive white Gaussian noise (AWGN) at the  $i$ -th port of the  $k$ -th CU and  $s_k$  represents the transmitted symbol dedicated to the  $k$ -th user. Without loss of generality, we assume the communication symbols have unit power, i.e.,  $\mathbb{E}[|s_k|^2] = 1$ . Then, without considering the noise due to the dominant multiuser interference, the average SIR received at the  $i$ -th port of the  $k$ -th user is given by

$$\Gamma_{i,k} = \frac{|h_k^{(i,k)}|^2}{\sum_{j=1, j \neq k}^{N_t} |h_k^{(i,j)}|^2}. \quad (3)$$

Therefore, the CU wishes to select the port to maximize the average SIR, i.e.,  $\max_{i \in [1, N_r]} \Gamma_{i,k}$ . However, the problem is challenging to solve, especially for a large  $N_r$ . This is because calculating the SIR needs all the CSI, whose system overhead is intolerable. This issue inevitably becomes more severe in mobile scenarios, where the channels vary quickly and thus more frequent CSI estimation is required.

## III. PORT SELECTION BASED ON BANDIT LEARNING

To address the above problem, we introduce a bandit-learning-based algorithm, enabling the users to iteratively observe and record the SIR at each port. Denoting  $l$  as the iteration number, the port selection decision will be made based on historical decisions and previously collected rewards. In this online learning process, the user attempts to iteratively learn the SIR received at each port and progressively improve the selection decisions. It is worth pointing out that the proposed learning-based selection approach does not need all the ports' CSI and the port selection can be updated automatically when the channel changes, which is more suitable for the dynamic channel environments, such as the high mobility scenarios.

### A. Problem Transformation

We formulate the port selection as a bandit learning process where the user, designed as a gambler/player, must decide which port, designed as an arm, to play in a sequence of trials to maximize the reward. Note that as we do not make any assumptions about the statistics of the channel state of each port, we model the process as a non-stochastic bandit problem and give the definitions as follows.

- $a[l]$  denotes the port selection decisions. The indicator function  $\mathbb{1}\{a[l] = i\} = 1$  indicates that the user selects the  $i$ -th port for receiving the signal at the  $l$ -th iteration.
- $\mathbf{Q}_A = \{q_1, \dots, q_{N_r}\}$  is the action space consisting of the available port.
- $p_i[l]$  denotes the possibility of choosing the  $i$ -th port at the  $l$ -th iteration.
- $\theta_i[l]$  is the reward receiving by choosing the  $i$ -th port at the  $l$ -th iteration.  $\Theta[l] = \{\theta_i[l]\}$  represents the reward space where  $\theta_a[l]$  is updated after the  $l$ -th iteration.<sup>1</sup>

<sup>1</sup>We assume the reward for each arm lies in  $[0, 1]$ , which is a standard assumption and widely used in bandit literature [10]. This assumption can be satisfied by normalizing the observed reward with respect to its maximum,  $\theta_{\max}$ , and its minimum,  $\theta_{\min}$ .

Based on the assumptions and definitions above, we transform the port selection problem into a bandit learning problem. The transformed problem for the  $k$ -th user at the  $l$ -th iteration is given by

$$\max_{a[l]} \Gamma_{a[l],k}[l], \quad (4a)$$

$$\text{s. t. } \sum_{i=1}^{N_r} \mathbb{1}\{a[l] = i\} = 1, \quad (4b)$$

$$q_{a[l]} \in \mathbf{Q}_A. \quad (4c)$$

where (4b) denotes that only one port can be activated at each iteration.

### B. Proposed Port Selection Algorithm

In the following, we elaborate on the proposed port selection algorithm, which consists of three phases, i.e., initialization, selection, and update. In each iteration, the user selects an option according to a specific distribution, i.e., the Gibbs distribution, which is a mixture of a uniform distribution  $\eta$  and an empirical performance-related distribution  $\lambda_i[l]$ . The uniform distribution ensures that the user keeps exploring occasionally and discovers the potential better options that were non-optimal previously.

- In the initialization phase, the weights for performance-related distribution at each port,  $\lambda_i[0]$  is set to  $\lambda_i[0] = 1$ , and all the rewards,  $\theta_i[0]$ , are set to zero.
- In the selection phase, for each  $q_i \in \mathbf{Q}_A$ , the user firstly observes the possibilities, which are calculated based on the weights as

$$p_i[l] = (1 - \eta) \frac{\lambda_i[l]}{\sum_{i=1}^{N_r} \lambda_i[l]} + \frac{\eta}{N_r}, \forall i \in [1, \dots, N_r], \quad (5)$$

where  $\eta$  is a tuning parameter to balance the trade-off between exploration and exploitation when the user is choosing which port to select. When  $\eta$  is small, the user tends to play the machine that will give the highest payout (referred to as “exploitation”). In such circumstances, other ports that have higher expected returns might be missed. On the other hand, if  $\eta$  is too large, the user will spend a lot of time trying out all the ports and collecting rewards (referred to as “exploration”), and may not play the machine with the highest reward frequently enough to maximize the objective. Additionally, the purpose of incorporating the uniform distribution during the mixing process is to ensure that the algorithm considers all possible actions and obtains accurate estimates of the rewards associated with each action. Without the uniform distribution, the algorithm may overlook a potentially beneficial action because initial observations of the rewards for this action may be low, leading to a failure to select the action and subsequently miss out on larger rewards that may occur later. For each iteration, the user draws the action  $a[l]$  and makes the decision as  $q[l] = q_{a[l]}$ .

### Algorithm 1 The Proposed Prediction-Bandit Algorithm

**Input:**  $\mathbf{Q}_A, \Theta[0]$

**Output:**  $\{a[l]\}_{l \in \mathcal{L}}$

**For each**  $l = 1, 2, \dots$

1: **Phase I:** Initialization

2: Initialize  $\lambda_i = 1, \forall i \in [1, \dots, N_r]$

3: **Phase II:** Selection

4: Set  $p_i[l] = (1 - \eta) \frac{\lambda_i[l]}{\sum_{i=1}^{N_r} \lambda_i[l]} + \frac{\eta}{N_r}, \forall i \in [1, \dots, N_r]$

5: **for**  $1, \dots, N_r$

6: Set  $\hat{p}_i[l] = p_i[l]$

7: Select  $a[l]$  randomly accordingly to the probabilities  $\hat{p}_1[l], \dots, \hat{p}_{N_r}[l]$

8: **Phase III:** Update

9: Receive reward  $\theta_{a[l]}[l]$  by (6)

10: **for**  $j = 1, \dots, N_r$

11: **if**  $j = a[l]$  **then**

12: Set  $\hat{\theta}_j[l] = \theta_j[l]/p_j[l]$

13: **else**

14: Set  $\hat{\theta}_j[l] = 0$

15: Set  $\lambda_j[l+1] = \lambda_j[l] e^{\eta \hat{\theta}_j[l]/N_r}$

- In the update phase, the user receives feedback of the selection of the last iteration and the reward can be calculated as

$$\theta_{a[l]}[l] = \frac{\left| h_k^{(a[l],i)}[l] \right|^2}{\sum_{j=1, j \neq k}^{N_t} \left| h_k^{(a[l],j)}[l] \right|^2}, \quad (6)$$

where the CSI at the  $a[l]$ -th port can be obtained using channel training.

For the drawn action  $a[l] = i$ , we set the estimated reward  $\hat{\theta}_i[l] = \theta_i[l]/p_i[l]$  which compensates the reward of actions are unlikely to be chosen. Then, the weights  $\lambda_i[l]$  can be correspondingly updated. The complete algorithm is given in Algorithm 1.

With the definition of the estimated reward, we have

$$\hat{\theta}_i[l] \leq 1/p_i[l] \leq N_r/\eta, \quad (7)$$

$$\sum_{i=1}^{N_r} p_i[l] \hat{\theta}_i[l] = p_{a[l]}[l] \cdot \frac{\theta_{a[l]}[l]}{p_{a[l]}[l]} = \theta_{a[l]}[l], \quad (8)$$

which can be initially derived and uses the fact that the probability of the port that is not chosen is equal to 0.

### C. Performance Analysis on Bandit-Learning Algorithm

In this subsection, we measure the algorithm performance by giving the lower bound of the rewards and the upper bound of worst-case regret. For any  $L \geq 0$ , we calculate cumulative reward at time  $L$ , given by  $\sum_{l=1}^L \theta_{a[l]}[l]$ , which is obtained by choosing the arm  $a[l]$  at the  $l$ -th iteration.

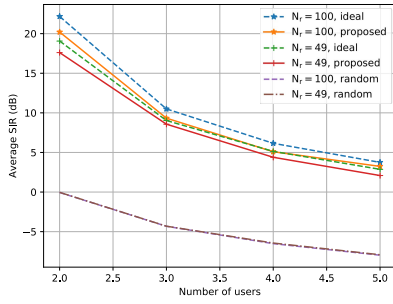


Fig. 2. Average SIR versus number of users under different  $N_r$ , with  $\eta = 0.07$ .

**Theorem 1:** By considering the expectations of the received reward, the expected reward is lower bounded as

$$\sum_{l=1}^L \theta_{a[l]}[l] \geq (1 - \eta) \sum_{l=1}^L \hat{\theta}_j[l] - \frac{N_r \ln N_r}{\eta} - (e - 2) \frac{\eta}{N_r} \sum_{l=1}^L \sum_{i=1}^{N_r} \hat{\theta}_k[l], \forall j. \quad (9)$$

*Proof:* see Appendix . ■

Then, we measure the algorithm performance by worst-case regret, defined as

$$\sum_{l=1}^L \theta_{\max}[l] - \mathbb{E} \left\{ \sum_{l=1}^L \theta_{a[l]}[l] \right\}, \quad (10)$$

where the first item denotes the reward at time  $L$  obtained by choosing the globally best decision at each iteration. Then, we have the following theorem that bounds the expected regret.

**Theorem 2:** For any  $N_r \geq 0$  and for any  $\eta \in (0, 1]$ , we have the upper bound of the expected regret as

$$\sum_{l=1}^L \theta_{\max}[l] - \mathbb{E} \left\{ \sum_{l=1}^L \theta_{a[l]}[l] \right\} \leq (e - 1) \eta \sum_{l=1}^L \theta_{\max}[l] + \frac{N_r \ln N_r}{\eta} \quad (11)$$

*Proof:* see Appendix . ■

#### IV. SIMULATION RESULTS

In this section, we provide numerical analysis to evaluate the performance of the proposed algorithms. We consider vehicular communications, where a BS serves multiple downlink vehicular users that are equipped with a fluid antenna, and the channel is time-varying due to the ever-changing channel environments. Unless stated otherwise, we assume that  $W_{r1} = W_{r2} = 2$ . Therefore, the inherent time-changing statistics require adaptively optimal port selection. To demonstrate the effectiveness of the proposed algorithm, we compare it with random selection with a uniform distribution as the baseline method. We also show the results of the ideal case where the CSI at each port is assumed to be perfectly known.

First, we study the average SIR of the proposed method with the comparison of the baseline and the ideal case. As shown in Fig. 2, it can be observed that the average SIR of the proposed method is very close to the ideal case, achieving over 90% of the average SIR of the ideal case when the number of users is 2. When the number of users increases, the average SIR decreases accordingly as there is more

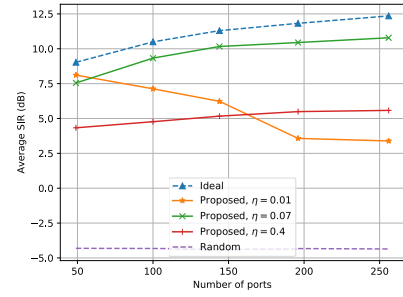


Fig. 3. Average SIR versus number of ports under different  $\eta$ , with the number of users set to 3.

interference. Nevertheless, the performance of the proposed method significantly outperforms the baseline method and keeps comparable performance with the ideal case. It can also be noted that the results for random selection in the case of  $N_r = 100$  and  $N_r = 49$  remain in a similar low level, indicating that the benefit for fluid antenna cannot be achieved without a carefully designed port selection algorithm.

Fig. 3 illustrates the average SIR of the proposed method under varying numbers of ports: 49 ( $7 \times 7$ ), 100 ( $10 \times 10$ ), 144 ( $12 \times 12$ ), and 256 ( $14 \times 14$ ). As the number of ports increases, both the average SIR of the ideal case and the average SIR of the proposed method with  $\eta = 0.07$  exhibit higher levels. However, for a smaller  $\eta$ , i.e.,  $\eta = 0.01$ , while it achieves superior performance at  $N_r = 49$ , its performance remarkably deteriorates as the number of ports increases. This outcome arises from the tendency of users to favor ports reflecting the highest SIR, potentially overlooking other valuable ports. On the other hand, with a larger  $\eta$ , i.e.,  $\eta = 0.4$ , the average SIR remains at a low level, as the user spends a lot of time trying out all the ports but may not frequently choose the port with the highest reward.

#### V. CONCLUSION

In this letter, we adopted multi-armed bandit theory to propose an online-learning induced port selection method for fluid antenna, without the need the CSI of all ports to calculate the instantaneous SIR. The proposed learning-induced port selection facilitated the implementation of  $s$ -FAMA and we also derived the analytical result of the upper bound of the expected regret which quantified the performance of our proposed algorithm. Finally, the simulation results confirmed its superior performance, which showed that the proposed algorithm can achieve much better performance than the random selection case and obtained almost the same performance as the ideal case where the port selection is based on all the ports' CSI.

#### APPENDIX A

Following [10], we denote the sum of the weights of all ports at the  $l$ -th iteration as  $\Lambda_l = \sum_{i=1}^{N_r} \lambda_i[l]$ . We have

$$\frac{\Lambda_{l+1}}{\Lambda_l} = \sum_{i=1}^{N_r} \frac{\lambda_i[l+1]}{\Lambda_l} \quad (12)$$

$$= \sum_{i=1}^{N_r} \frac{\lambda_i[l]}{\Lambda_l} e^{\frac{\eta}{N_r} \hat{\theta}_i[l]} \quad (13)$$



$$= \frac{1}{1-\eta} \sum_{i=1}^{N_r} \left( p_i[l] - \frac{\eta}{N_r} \right) e^{\frac{\eta}{N_r} \hat{\theta}_i[l]} \quad (14)$$

$$\stackrel{(a)}{\leq} \frac{\sum_{i=1}^{N_r} \left( p_i[l] - \frac{\eta}{N_r} \right)}{1-\eta} \left( 1 + \frac{\eta \hat{\theta}_i[l]}{N_r} + (e-2) \left( \frac{\eta \hat{\theta}_i[l]}{N_r} \right)^2 \right) \quad (15)$$

$$\stackrel{(b)}{\leq} 1 + \frac{\eta \sum_{i=1}^{N_r} p_i[l] \hat{\theta}_i[l]}{N_r(1-\eta)} + (e-2) \frac{\eta^2 \sum_{i=1}^{N_r} p_i[l] (\hat{\theta}_i[l])^2}{N_r^2(1-\eta)} \quad (16)$$

where the inequality (a) holds as  $e^x \leq 1+x+(e-2)x^2$  for  $x \leq 1$  and (b) holds by combining (5). Recalling  $\sum_{i=1}^{N_r} p_i[l] \hat{\theta}_i[l] = \theta_{a[l]}[l]$ , we have

$$p_i[l] (\hat{\theta}_i[l])^2 = \theta_{a[l]}[l] \hat{\theta}_i[l] \leq \hat{\theta}_i[l] = \sum_{i=1}^{N_r} \hat{\theta}_i[l]. \quad (17)$$

Then, we can derive

$$\frac{\Lambda_{l+1}}{\Lambda_l} \leq 1 + \frac{\eta \theta_{a[l]}[l]}{N_r(1-\eta)} + (e-2) \frac{\eta^2 \sum_{i=1}^{N_r} \hat{\theta}_i[l]}{N_r^2(1-\eta)}. \quad (18)$$

Taking logarithms for the both sides of (18) and utilizing the fact that  $1+x \leq e^x$ , we have

$$\ln \frac{\Lambda_{l+1}}{\Lambda_l} \leq \frac{\eta \theta_{a[l]}[l]}{N_r(1-\eta)} + (e-2) \frac{\eta^2 \sum_{i=1}^{N_r} \hat{\theta}_i[l]}{N_r^2(1-\eta)}. \quad (19)$$

For both sides of (19), we summing over  $l \in [1, \dots, L]$ , we have

$$\begin{aligned} \ln \frac{\Lambda_{L+1}}{\Lambda_1} &\leq \frac{\eta \sum_{l=1}^L \theta_{a[l]}[l]}{N_r(1-\eta)} \\ &\quad + (e-2) \frac{\eta^2 \sum_{l=1}^L \sum_{i=1}^{N_r} \hat{\theta}_i[l]}{N_r^2(1-\eta)}. \end{aligned} \quad (20)$$

For any port option  $j$ , it can be immediately derived that

$$\ln \frac{\Lambda_{L+1}}{\Lambda_1} \leq \ln \frac{\lambda_j[L+1]}{\Lambda_1}. \quad (21)$$

Considering

$$\lambda_j[L+1] = \lambda_j[L] e^{\eta \hat{\theta}_j[L]/N_r} \quad (22)$$

$$= \prod_{l=1}^L e^{\eta \hat{\theta}_j[l]/N_r} = e^{\eta \sum_{l=1}^L \hat{\theta}_j[l]/N_r}, \quad (23)$$

we have

$$\ln \frac{\lambda_j[L+1]}{\Lambda_1} = \frac{\eta}{N_r} \sum_{l=1}^L \hat{\theta}_j[l] - \ln N_r. \quad (24)$$

Combining with (20), we get

$$\begin{aligned} \sum_{l=1}^L \theta_{a[l]}[l] &\geq (1-\eta) \sum_{l=1}^L \hat{\theta}_j[l] - \frac{N_r \ln N_r}{\eta} \\ &\quad - (e-2) \frac{\eta}{N_r} \sum_{l=1}^L \sum_{i=1}^{N_r} \hat{\theta}_i[l], \forall j. \end{aligned} \quad (25)$$

which completes the proof.

## APPENDIX B

We first take the expectation of each estimated reward with respect to the distribution of  $\langle a[1], \dots, a[L] \rangle$  as

$$\begin{aligned} &\mathbb{E} \left[ \hat{\theta}_i[l] \middle| a[1], \dots, a[L] \right] \\ &= \mathbb{E} \left[ p_i[l] \cdot \frac{\theta_i[l]}{p_i[l]} + (1-p_i[l]) \cdot 0 \right] = \theta_i[l] \end{aligned} \quad (26)$$

Combining (11) and (26), we have

$$\begin{aligned} \mathbb{E} \left\{ \sum_{l=1}^L \theta_{a[l]}[l] \right\} &\geq (1-\eta) \sum_{l=1}^L \theta_j[l] \\ &\quad - \frac{N_r \ln N_r}{\eta} - (e-2) \frac{\eta}{N_r} \sum_{l=1}^L \sum_{i=1}^{N_r} \theta_i[l] \end{aligned} \quad (27)$$

Since  $j$  was chosen arbitrarily and

$$\sum_{l=1}^L \sum_{i=1}^{N_r} \theta_i[l] \leq N_r \sum_{l=1}^L \theta_{\max}[l]. \quad (28)$$

Combining (28) and (27) yields (11), which completes the proof.

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